Igor L. R. Azevedo

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Experience

The University of Tokyo — Tokyo, Japan
Research Assistant — Apr 2023 – Apr 2025

***Nikkei** (Financial Times) collaboration – news recommendation* Aug 2023 – Apr 2025

- Built multi-agent LLM pipelines with agentic retrieval and RAG using AWS Bedrock, LangChain, Strands SDK, and Neo4j to mine user behaviour patterns
- Built data pipelines and feature engineering over Nikkei user logs to train news recommenders
- Modelled implicit negative feedback with candidate-aware attention to reduce popularity bias
- Built NewsReX, a PyTorch/JAX framework for reproducible recommendation, 2.8x faster training

***Toyota** collaboration – mobility and point-of-interest recommendation* Apr 2023 – Aug 2023

- Processed vehicle GPS telemetry into trajectories and user–place graphs to predict driver destinations
- Researched graph-attention point-of-interest recommendation for in-car multimedia systems
- Built an automated evaluation system in PyTorch that ran and compared models on driving logs

VOGA, acquired by BTG Pactual — Brasília, Brazil
Development Team Lead — July 2021 – Apr 2023

- Architected a real-time portfolio data platform monitoring over USD 300M in client assets
- Built AWS infrastructure as code with Terraform, Docker and Jenkins CI/CD
- Automated reporting workflows into ETL pipelines and self-service dashboards
- Led 7 engineers through the integration into **BTG Pactual**, Latin America's largest investment bank

Cellcrypt, secure communications — London (Remote)
Applied Machine Learning Intern — Sept 2020 – July 2021

- Built an ML training and inference pipeline on AWS that tuned VoIP codecs, lifting call quality 7%
- Diagnosed call degradation from packet captures and built a C++ VoIP testbed (PJSIP)

Education

Imperial College London — London, United Kingdom
Master of Research in Artificial Intelligence and Machine Learning — Sept 2025 – Sept 2026

- Researched multimodal representation collapse in contrastive learning and mixture of experts
- Investigated expert collapse in mixture-of-experts routing when minority classes are scarce

University of Brasília — Brasília, Brazil
BSc in Electrical Engineering, GPA 4.35/5.0 — Aug 2016 – June 2022

- Researched FPGA hardware acceleration of SHA-3 cryptography for IoT devices
- Applied machine learning to financial market prediction in the undergraduate thesis

Papers and Projects

NewsReX, open-source PyTorch/JAX framework for news recommendation — **CIKM 2026**
AWRS, avoidance-aware news recommender — **SIAM SDM 2025**
learning-ai.cafe, a collaborative open-source AI textbook, from intuition to depth

Honours and Awards

Japanese Government (MEXT) Research Scholarship, around 9% global acceptance — 2023 – 2025
PIBIC/CNPq Research Scholarship, top 4% of undergraduates in Brazil — 2019 – 2020
SIAM Travel Award, SIAM SDM 2025 — 2025
Y Combinator Startup School, Travel Grant — 2026
3rd Place, HackTheLaw, University of Cambridge and Stanford CodeX — 2026

Skills

- Technical: Python, SQL, C++, Java, PyTorch, JAX, PyTorch Geometric, scikit-learn, Strands Agents, RAG, AWS (ECS, EC2, RDS, Bedrock, Lambda), Claude Code, Agile delivery, Docker
- Languages: Portuguese (native), English (fluent), Spanish (intermediate), Japanese (basic)